

监控与告警方案

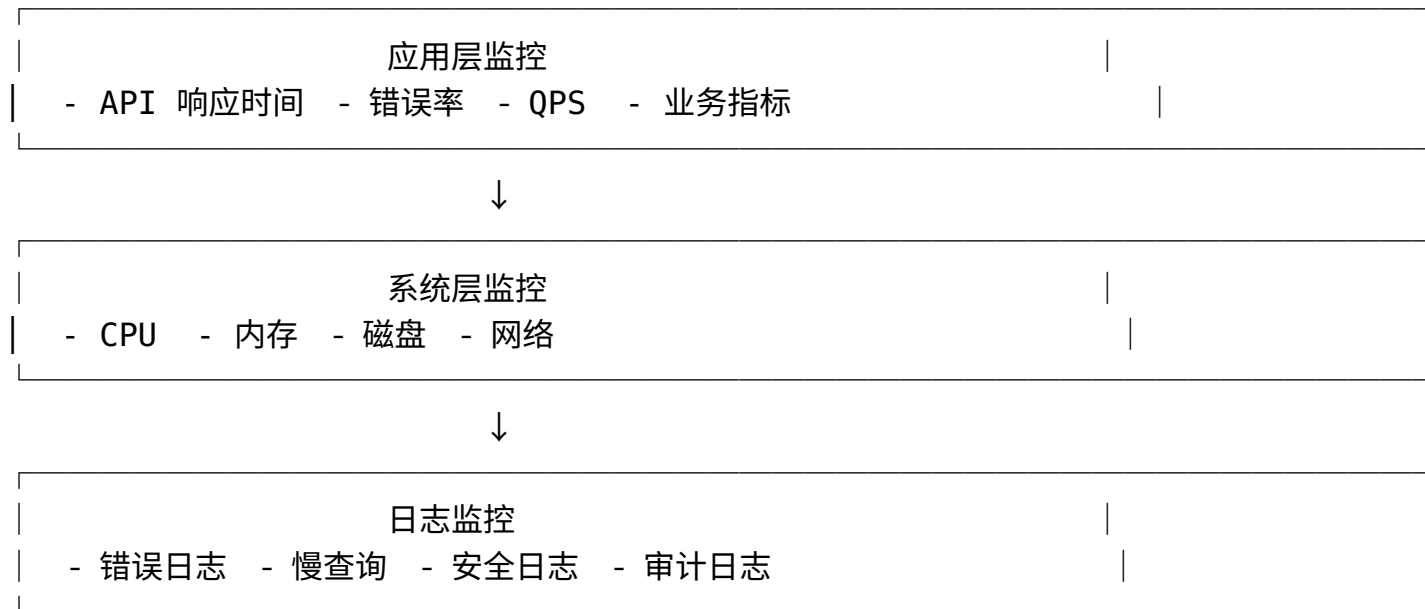
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一、监控架构

1.1 监控层次



1.2 监控工具

层级	工具	用途
应用监控	Prometheus + Grafana	指标收集与可视化
日志监控	Loki + Grafana	日志聚合与查询
告警	Alertmanager	告警通知
链路追踪	Jaeger (可选)	请求链路追踪

二、应用监控

2.1 核心指标

指标	说明	告警阈值
API 响应时间	P95	< 500ms >1s
API 错误率	< 1%	>5%
QPS	根据容量规划	突增 200%
活跃用户数	业务指标	-
案件创建数	业务指标	-

2.2 Rust 指标收集

```
// src/metrics.rs
use prometheus::{Registry, Counter, Gauge, Histogram, register_counter, re

pub struct AppMetrics {
    pub registry: Registry,
    pub http_requests_total: Counter,
    pub http_request_duration: Histogram,
    pub active_users: Gauge,
    pub case_count: Gauge,
    pub file_upload_total: Counter,
}

impl AppMetrics {
    pub fn new() -> Result<Self, prometheus::Error> {
        let registry = Registry::new();

        let http_requests_total = register_counter!(
            "http_requests_total",
            "Total HTTP requests",
            &["method", "path", "status"],
            registry
        )?;

        let http_request_duration = register_histogram!(
            "http_request_duration_seconds",
            "HTTP request duration in seconds",
```

```

        vec![0.01, 0.05, 0.1, 0.25, 0.5, 1.0, 2.5, 5.0],
        registry
    )?;

    let active_users = register_gauge!(
        "active_users",
        "Number of active users",
        registry
    )?;

    let case_count = register_gauge!(
        "case_count",
        "Total number of cases",
        registry
    )?;

    let file_upload_total = register_counter!(
        "file_upload_total",
        "Total file uploads",
        registry
    )?;

    Ok(Self {
        registry,
        http_requests_total,
        http_request_duration,
        active_users,
        case_count,
        file_upload_total,
    })
}

pub fn record_request(&self, method: &str, path: &str, status: u16, duration: f64) {
    self.http_requests_total
        .with_label_values(&[method, path, &status.to_string()])
        .inc();

    self.http_request_duration.observe(duration);
}
}

```

```
// 使用示例
#[middleware]
pub async fn metrics_middleware(
    State(metrics): State<Arc<AppMetrics>>,
    request: Request<Body>,
    next: Next,
) -> Response {
    let start = Instant::now();
    let method = request.method().to_string();
    let path = request.uri().path().to_string();

    let response = next.run(request).await;
    let status = response.status().as_u16();
    let duration = start.elapsed().as_secs_f64();

    metrics.record_request(&method, &path, status, duration);

    response
}
```

2.3 指标导出接口

```
// src/routers/metrics.rs
use axum::response::IntoResponse;
use prometheus::TextEncoder;

pub async fn metrics_handler(
    State(metrics): State<Arc<AppMetrics>>,
) -> impl IntoResponse {
    let encoder = TextEncoder::new();
    let metric_families = metrics.registry.gather();

    let mut buffer = Vec::new();
    encoder.encode(&metric_families, &mut buffer).unwrap();

    (
        [(http::header::CONTENT_TYPE, TextEncoder::FORMAT_TYPE)],
        buffer,
    )
}
```

```
}
```

```
// 路由配置
// GET /metrics
```

三、系统监控

3.1 Node Exporter 配置

```
# prometheus/node-exporter.yml
scrape_configs:
  - job_name: 'node'
    static_configs:
      - targets: ['localhost:9100']
```

3.2 系统指标

指标	说明	告警阈值
CPU 使用率	服务器 CPU 使用	>80%
内存使用率	服务器内存使用	>85%
磁盘使用率	磁盘空间使用	>80%
磁盘 IO	磁盘读写延迟	>100ms
网络流量	网络出入流量	突增 200%

3.3 进程监控

```
#!/bin/bash
# monitor_process.sh

SERVICE="legal-minds"
ALERT_THRESHOLD=3 # 连续 3 次失败则告警

check_count=0

while true; do
    if systemctl is-active --quiet ${SERVICE}; then
        check_count=0
    else
        check_count=$((check_count + 1))
```

```
        if [ ${check_count} -ge ${ALERT_THRESHOLD} ]; then
            echo "[$(date)] 告警: 服务 ${SERVICE} 已停止" | mail -s "服务告警" a
            # 尝试重启
            systemctl start ${SERVICE}
            check_count=0
        fi
    fi

    sleep 30
done
```

四、日志监控

4.1 日志级别

```
// src/logging.rs
use tracing::{error, warn, info, debug, trace};

// 错误日志 - 需要立即处理
error!("数据库连接失败: {}", error);

// 警告日志 - 需要注意
warn!("API 响应时间超过阈值: {}ms", duration);

// 信息日志 - 正常运行信息
info!("用户登录: user_id={}", user_id);

// 调试日志 - 开发调试用
debug!("SQL 查询: {}", query);

// 追踪日志 - 详细追踪
trace!("请求头: {:?}", headers);
```

4.2 日志收集配置

```
# loki/config.yml
auth_enabled: false

server:
    http_listen_port: 3100
```

```
common:
  path_prefix: /loki
  replication_factor: 1
  storage:
    filesystem:
      chunks_directory: /loki/chunks
      rules_directory: /loki/rules

schema_config:
  configs:
    - from: 2026-01-01
      store: boltdb-shipper
      object_store: filesystem
      schema: v11
      index:
        prefix: index_
        period: 24h

limits_config:
  retention_period: 2160h # 90 天

ruler:
  alertmanager_url: http://localhost:9093
```

4.3 日志查询示例

```
// 错误日志统计
sum(rate({app="legal-minds", level="error"}[5m])) by (service)

// 慢查询日志
{app="legal-minds"} |= "slow query" | duration > 1s

// 登录失败统计
sum(rate({app="legal-minds"} |= "login_failed"[5m])) by (user_id)
```

五、告警规则

5.1 Prometheus 告警规则

```
# prometheus/alerts.yml
groups:
  - name: legal-minds-alerts
    rules:
      # API 错误率过高
      - alert: HighErrorRate
        expr: sum(rate(http_requests_total{status=~"5.."}[5m])) / sum(rate(http_requests_total{status!="5.."}[5m])) > 0.05
        for: 5m
        labels:
          severity: critical
        annotations:
          summary: "API 错误率过高"
          description: "当前错误率: {{ $value | humanizePercentage }}"

      # API 响应时间过长
      - alert: HighLatency
        expr: histogram_quantile(0.95, sum(rate(http_request_duration_seconds)[5m])) > 0.5
        for: 5m
        labels:
          severity: warning
        annotations:
          summary: "API 响应时间过长"
          description: "P95 延迟: {{ $value | humanizeDuration }}"

      # 服务宕机
      - alert: ServiceDown
        expr: up{job="legal-minds"} == 0
        for: 1m
        labels:
          severity: critical
        annotations:
          summary: "服务宕机"
          description: "实例 {{ $labels.instance }} 已宕机"

      # CPU 使用率过高
      - alert: HighCPUUsage
```



```

    expr: 100 - (avg by(instance) (irate(node_cpu_seconds_total{mode="idle"}))
    for: 10m
    labels:
      severity: warning
    annotations:
      summary: "CPU 使用率过高"
      description: "CPU 使用率: {{ $value | humanizePercentage }}"

# 内存使用率过高
- alert: HighMemoryUsage
  expr: (1 - (node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes))
  for: 10m
  labels:
    severity: warning
  annotations:
    summary: "内存使用率过高"
    description: "内存使用率: {{ $value | humanizePercentage }}"

# 磁盘空间不足
- alert: LowDiskSpace
  expr: (1 - (node_filesystem_avail_bytes / node_filesystem_size_bytes))
  for: 10m
  labels:
    severity: warning
  annotations:
    summary: "磁盘空间不足"
    description: "磁盘使用率: {{ $value | humanizePercentage }}"

```

5.2 Alertmanager 配置

```

# alertmanager/config.yml
global:
  resolve_timeout: 5m
  smtp_smarthost: 'smtp.example.com:587'
  smtp_from: 'alert@example.com'
  smtp_auth_username: 'alert@example.com'
  smtp_auth_password: 'password'

route:
  group_by: ['alertname']

```

```
group_wait: 10s
group_interval: 10s
repeat_interval: 1h
receiver: 'email-notifications'
routes:
  - match:
      severity: critical
      receiver: 'sms-notifications'
      repeat_interval: 15m

receivers:
  - name: 'email-notifications'
    email_configs:
      - to: 'admin@example.com'
        send_resolved: true

  - name: 'sms-notifications'
    webhook_configs:
      - url: 'http://sms-gateway.example.com/alert'

inhibit_rules:
  - source_match:
      severity: 'critical'
    target_match:
      severity: 'warning'
    equal: ['alertname', 'instance']
```

六、告警通知

6.1 通知渠道

渠道	适用场景	配置
邮件	一般告警	SMTP
短信	紧急告警	短信网关 API
钉钉	团队通知	Webhook
企业微信	团队通知	Webhook
Slack	团队通知	Webhook

6.2 钉钉通知配置

```
# alertmanager/config.yml
receivers:
  - name: 'dingtalk'
    webhook_configs:
      - url: 'https://oapi.dingtalk.com/robot/send?access_token=YOUR_TOKEN'
        send_resolved: true
```

6.3 告警模板

```
{{ define "dingtalk.message" }}
## {{ .Status | toUpper }}
### {{ .CommonAnnotations.summary }}
{{ .CommonAnnotations.description }}
```

****告警详情:****

```
{{ range .Alerts }}
- 实例: {{ .Labels.instance }}
- 开始时间: {{ .StartsAt }}
{{ end }}
```

****告警链接:**** {{ .ExternalURL }}

```
{{ end }}
```

七、监控面板

7.1 Grafana 面板配置

```
{
  "dashboard": {
    "title": "LegalMinds 监控",
    "panels": [
      {
        "title": "API 请求量",
        "type": "graph",
        "targets": [
          {
            "expr": "sum(rate(http_requests_total[5m])) by (method)",
            "legendFormat": "{{method}}"
          }
        ]
      }
    ]
  }
}
```

```

    }
  ]
},
{
  "title": "API 响应时间 (P95)",
  "type": "graph",
  "targets": [
    {
      "expr": "histogram_quantile(0.95, sum(rate(http_request_duration_microseconds{instance=\"$instance\"}[5m])) by (le))",
    }
  ]
},
{
  "title": "错误率",
  "type": "graph",
  "targets": [
    {
      "expr": "sum(rate(http_requests_total{status=~\"5..\"}[5m])) / sum(rate(http_requests_total[5m]))",
    }
  ]
},
{
  "title": "系统资源",
  "type": "row",
  "panels": [
    {
      "title": "CPU 使用率",
      "type": "gauge",
      "targets": [
        {
          "expr": "100 - (avg(irate(node_cpu_seconds_total{mode=\"idle\"}[5m])) * 100)",
        }
      ]
    },
    {
      "title": "内存使用率",
      "type": "gauge",
      "targets": [
        {
          "expr": "(1 - (node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes)) * 100",
        }
      ]
    }
  ]
}

```

```
        ]
      }
    ]
  }
]
}
```

7.2 业务指标面板

指标	查询	说明
今日案件创建数	<code>increase(case_count[24h])</code>	每日新增案件
今日文件上传数	<code>increase(file_upload_total[24h])</code>	每日上传文件
活跃用户数	<code>active_users</code>	当前在线用户
搜索次数	<code>increase(search_requests_total[24h])</code>	每日搜索量

八、部署配置

8.1 Docker Compose

```
# docker-compose.monitoring.yml
version: '3.8'

services:
  prometheus:
    image: prom/prometheus:latest
    ports:
      - "9090:9090"
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml
      - prometheus_data:/prometheus
    command:
      - '--config.file=/etc/prometheus/prometheus.yml'
      - '--storage.tsdb.path=/prometheus'
      - '--storage.tsdb.retention.time=15d'

  grafana:
    image: grafana/grafana:latest
    ports:
```

```
- "3000:3000"
volumes:
  - grafana_data:/var/lib/grafana
  - ./grafana/provisioning:/etc/grafana/provisioning
environment:
  - GF_SECURITY_ADMIN_PASSWORD=admin
```

```
loki:
  image: grafana/loki:latest
  ports:
    - "3100:3100"
  volumes:
    - ./loki.yml:/etc/loki/local-config.yaml
    - loki_data:/loki
```

```
alertmanager:
  image: prom/alertmanager:latest
  ports:
    - "9093:9093"
  volumes:
    - ./alertmanager.yml:/etc/alertmanager/alertmanager.yml
```

```
volumes:
  prometheus_data:
  grafana_data:
  loki_data:
```

8.2 启动监控

启动监控栈

```
docker-compose -f docker-compose.monitoring.yml up -d
```

检查状态

```
docker-compose -f docker-compose.monitoring.yml ps
```

查看日志

```
docker-compose -f docker-compose.monitoring.yml logs -f prometheus
```

九、总结

9.1 监控清单

监控项	工具	状态
API 指标	Prometheus	
系统指标	Node Exporter	
日志聚合	Loki	
告警通知	Alertmanager	
可视化	Grafana	

9.2 告警级别

级别	说明	通知方式	响应时间
Critical	服务宕机、数据丢失	短信 + 邮件	5 分钟
Warning	性能下降、资源不足	邮件 + 钉钉	30 分钟
Info	常规通知	邮件	24 小时

9.3 联系人

角色	姓名	电话	邮箱
技术负责人	王舟	15378391447	main@mails.wedevs.org
运维负责人	[待填写]	[待填写]	[待填写]
紧急联系	[待填写]	[待填写]	[待填写]

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